Type MS · What encryption is and why we use it

Answer 1 of 5 · Question 3 part b · 1 mark

2020 NOV · Paper 02 · extract 1 of 6 · find part b in this mark scheme extract

| | | | | |
|---|---|---|---|---|
| 3 (a) | A1 | Award one mark for any of:<br>• Conversion of plain text into cipher (1)<br>• Converting information/data into a code (1)<br>• Converting information/data into an unreadable format (1) | Accept alternative wording | (1) |

| Question | mp | Answer | Additional Guidance | Mark |
|---|---|---|---|---|

Type MS · What encryption is and why we use it

Answer 1 of 5 · Question 3 part b · 1 mark

2020 NOV · Paper 02 · extract 2 of 6 · find part b in this mark scheme extract

| | | | | |
|---|---|---|---|---|
| 3 (b) | B1 | Award one mark for any of:<br>• To ensure that the data can only be read by an authorised person / can't be read by an unauthorised person. (1) | Accept alternative similar wording.<br>Do not accept hacking. | (1) |

| Question | mp | Answer | Additional Guidance | Mark |
|---|---|---|---|---|

Type MS · What encryption is and why we use it

Answer 1 of 5 · Question 3 part b · 1 mark

2020 NOV · Paper 02 · extract 3 of 6 · find part b in this mark scheme extract

| | | | | |
|---|---|---|---|---|
| 3 (c) (i) | C1<br>C2<br>C3<br>C4 | Award 4 marks for a correct response:<br><br>• CAL\|OTTA\|MUIN\|PO<br>• PO\|MUIN\|OTTA\|CAL<br><br>Award 1 mark each up to a maximum of 3 for:<br><br>• 4 lines of text (1)<br>• Zigzag arrangement of letters (1)<br>• Letters from each line reproduced / consistent use of the key 4: GSQTYXEXMSREP/ YKILQPWPEKJWH (1) | | (4) |

| Question | mp | Answer | Additional Guidance | Mark |
|---|---|---|---|---|

## Type MS · What encryption is and why we use it

### Answer 1 of 5 · Question 3 part b · 1 mark

2020 NOV · Paper 02 · extract 4 of 6 · find part b in this mark scheme extract

| | | | | |
|---|---|---|---|---|
| 3 (c) (ii) | C5<br>C6 | Award 2 marks for a linked explanation such as:<br><br>• Limited number of usable keys (1) to allow for sufficient movement of characters / so can easily be decoded by trial | | (2) |

5 of 16

## Type MS · What encryption is and why we use it

### Answer 1 of 5 · Question 3 part b · 1 mark

2020 NOV · Paper 02 · extract 5 of 6 · find part b in this mark scheme extract

| | | | | |
|---|---|---|---|---|
| | | and error / easy to use brute force to crack (1) | | |

## Type MS · What encryption is and why we use it

### Answer 1 of 5 · Question 3 part b · 1 mark

2020 NOV · Paper 02 · extract 6 of 6 · find part b in this mark scheme extract

| Question | mp | ref | Answer | Additional Guidance | Mark |
|---|---|---|---|---|---|

Type MS · What encryption is and why we use it

Answer 2 of 5 · Question 6 part a.iii · 1 mark

2019 JUN · Paper 01 · extract 1 of 5 · find part a.iii in this mark scheme extract

| | | | |
|---|---|---|---|
| 6(a)(i) | Features such as power consumption / processor / memory / number and type of components / physical size (1) can be optimised / chosen to fit just one specific task. (1) | | 2 |

| Question Number | Answer | Additional Guidance | Mark |
|---|---|---|---|
| 6(a)(ii) | Any **two** from:<br>• Printer (1)<br>• Bank card (magnetic strip) reader/scanner (1)<br>• Bank card contactless/NFC reader/scanner (1)<br>• Cash input and giving change device (1) | Allow any hardware that relates to paying by cash for MP4 | 2 |

| Question Number | Answer | Additional Guidance | Mark |
|---|---|---|---|
| 6(a)(iii) | To prevent unauthorised people from reading / understanding the bank card information (1) | Do not award 'to keep it from being hacked / accessed'. | 1 |

## Type MS · What encryption is and why we use it

### Answer 2 of 5 · Question 6 part a.iii · 1 mark

2019 JUN · Paper 01 · extract 2 of 5 · find part a.iii in this mark scheme extract

| Question Number | Answer | Additional Guidance | Mark |
|---|---|---|---|
| 6(b) | Award **one** mark for each of (maximum of 4):<br><br>1. Readability: high-level language is easily readable by a human because it is designed to look like English whereas a low-level language is binary patterns / mnemonics / abbreviations. (1)<br>2. Portability: code written in a high-level language is easily transportable across machines whereas a low-level language is designed for a specific microprocessor / piece of hardware. (1)<br>3. Uses: low-level languages are commonly used to write programs for hardware or devices, like printers, whereas high-level languages are commonly used for to write higher-level applications, such as word processors. (1)<br>4. Optimisation: low-level languages are used to improve performance, especially of hardware, whereas high-level languages have to go through multiple layers of software to get to the hardware, e.g. the need for HLL to be compiled/interpreted makes programmes run slower than LLL, HLL has a higher level of abstraction removing unnecessary detail (one to many), LLL keeps more of a detail (one to one) (1)<br>5. Ease of use: high-level languages usually have tools, functions, libraries and development environments which are accessible, even by those with little knowledge, whereas, low-level languages are often challenging to work with and require expertise and depth of knowledge. (1) | For each mark point:<br>• the general feature is followed by exemplification<br>• needs to be a comparison, which may be implicit | 4 |

## Type MS · What encryption is and why we use it

### Answer 2 of 5 · Question 6 part a.iii · 1 mark

2019 JUN · Paper 01 · extract 3 of 5 · find part a.iii in this mark scheme extract

| Question Number | Answer | Mark |
|---|---|---|
| 6(c) | Indicative content:<br><br>Characteristics – Any of the following:<br>• Implemented in software.<br>• Could involve the use of neural networks / clustering / modelling<br>• Algorithms are trained by exposure to data that has been validated / checked<br>• Algorithms learn by identifying commonalities between the validated / checked data<br>• Algorithms make predictions for new data based on prior learning<br>• Algorithms may incorporate recursive self-improvement<br><br>Uses – Any of the following:<br>• Game playing (opponents)<br>• Analytics (analyse buying patterns, predicting behaviours, predictive text, targeted advertising, personalised news feeds, friend suggestions)<br>• Image / Audio processing (recognising objects/patterns, face recognition)<br>• Logistics (scheduling, order fulfilment)<br>• Control systems (cars, manufacturing, weapons, navigation<br>• Diagnostic systems (medical, mechanical, electrical)<br>• Natural languages processing (chatterbots, chatbots, speech recognition, personal assistants)<br>• Robotics (dangerous situations, help aged or disabled)<br><br>Ethical issues – Any of the following:<br>• Training sets could include data that should not be used to make decisions (race, religion, gender, etc.) that could cause biased outcomes<br>• Is a computer to be trusted to make decisions (life-death)?<br>• If a computer discovers something that humans can't prove, should it be accepted as truth?<br>• How do we know that the answer produced is correct?<br>• Will people be comfortable interacting with machines that are considered intelligent?<br>• Some jobs may be at risk, especially those requiring human skills such as empathy or creativity. | 6 |

Type MS · What encryption is and why we use it

Answer 2 of 5 · Question 6 part a.iii · 1 mark

2019 JUN · Paper 01 · extract 4 of 5 · find part a.iii in this mark scheme extract

| Level | Mark | Descriptor |
|---|---|---|
| | 0 | No rewardable content. |
| Level 1 | 1–2 | Basic, independent points are made showing elements of knowledge and understanding of key concepts/principles of computer science.<br><br>The discussion will contain basic information with little linkage between points made. |
| Level 2 | 3-4 | Demonstrates adequate knowledge and understanding of key concepts/principles of computer science.<br><br>The discussion shows some linkages and lines of reasoning with some structure. |
| Level 3 | 5-6 | Demonstrates comprehensive knowledge and understanding by selecting relevant knowledge and understanding of key concepts/principles of computer science to support the discussion being presented.<br><br>The discussion shows a well-developed, sustained line of reasoning which is clear, coherent and logically structured. |

## Type MS · What encryption is and why we use it

### Answer 2 of 5 · Question 6 part a.iii · 1 mark

2019 JUN · Paper 01 · extract 5 of 5 · find part a.iii in this mark scheme extract

Pearson Education Limited. Registered company number 872828
with its registered office at 80 Strand, London, WC2R 0RL, United Kingdom

Type MS · What encryption is and why we use it

Answer 3 of 5 · Question 3 part a · 1 mark

2020 NOV · Paper 02 · extract 1 of 6 · find part a in this mark scheme extract

| | | | | |
|---|---|---|---|---|
| 3 (a) | A1 | Award one mark for any of:<br>• Conversion of plain text into cipher (1)<br>• Converting information/data into a code (1)<br>• Converting information/data into an unreadable format (1) | Accept alternative wording | (1) |

| Question | mp | Answer | Additional Guidance | Mark |
|---|---|---|---|---|

Type MS · What encryption is and why we use it

Answer 3 of 5 · Question 3 part a · 1 mark

2020 NOV · Paper 02 · extract 2 of 6 · find part a in this mark scheme extract

| | | | | |
|---|---|---|---|---|
| 3 (b) | B1 | Award one mark for any of:<br>• To ensure that the data can only be read by an authorised person / can't be read by an unauthorised person. (1) | Accept alternative similar wording.<br>Do not accept hacking. | (1) |

| Question | mp | Answer | Additional Guidance | Mark |
|---|---|---|---|---|

Type MS · What encryption is and why we use it

Answer 3 of 5 · Question 3 part a · 1 mark

2020 NOV · Paper 02 · extract 3 of 6 · find part a in this mark scheme extract

| | | | | |
|---|---|---|---|---|
| 3 (c) (i) | C1<br>C2<br>C3<br>C4 | Award 4 marks for a correct response:<br><br>• CAL\|OTTA\|MUIN\|PO<br>• PO\|MUIN\|OTTA\|CAL<br><br>Award 1 mark each up to a maximum of 3 for:<br><br>• 4 lines of text (1)<br>• Zigzag arrangement of letters (1)<br>• Letters from each line reproduced / consistent use of the key 4: GSQTYXEXMSREP/ YKILQPWPEKJWH (1) | | (4) |

| Question | mp | Answer | Additional Guidance | Mark |
|---|---|---|---|---|

Type MS · What encryption is and why we use it

Answer 3 of 5 · Question 3 part a · 1 mark

2020 NOV · Paper 02 · extract 4 of 6 · find part a in this mark scheme extract

| | | | | |
|---|---|---|---|---|
| 3 (c) (ii) | C5<br>C6 | Award 2 marks for a linked explanation such as:<br><br>• Limited number of usable keys (1) to allow for sufficient movement of characters / so can easily be decoded by trial | | (2) |

5 of 16

## Type MS · What encryption is and why we use it

### Answer 3 of 5 · Question 3 part a · 1 mark

2020 NOV · Paper 02 · extract 5 of 6 · find part a in this mark scheme extract

| | | | | |
|---|---|---|---|---|
| | | and error / easy to use brute force to crack (1) | | |



Match each answer number to the type practice PDF. Original exam material © Pearson Education Ltd.

## Type MS · What encryption is and why we use it

### Answer 3 of 5 · Question 3 part a · 1 mark

2020 NOV · Paper 02 · extract 6 of 6 · find part a in this mark scheme extract

| Question | mp | ref | Answer | Additional Guidance | Mark |
|---|---|---|---|---|---|

## Type MS · What encryption is and why we use it

### Answer 4 of 5 · Question 4 part a · 1 mark

2022 JUN · Paper 02 · extract 1 of 6 · find part a in this mark scheme extract

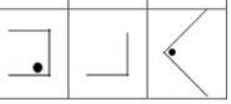
| | | | | |
|---|---|---|---|---|
| 4 (a) | A1 | **The only correct answer is B**<br><br>A is not correct because this is decryption<br>C is not correct because this is not an encryption<br>D is not correct because this is not an encryption | | (1) |

| Question | mp | Answer | Additional Guidance | Mark |
|---|---|---|---|---|

## Type MS · What encryption is and why we use it

### Answer 4 of 5 · Question 4 part a · 1 mark

2022 JUN · Paper 02 · extract 2 of 6 · find part a in this mark scheme extract

| | | | | |
|---|---|---|---|---|
| 4 (b) | B1<br>B2<br>B3 | **Award 1 mark for each correct symbol**<br><br>**Word** M A Y<br>**Symbol**  | | (3) |

| Question | mp | Answer | Additional Guidance | Mark |
|---|---|---|---|---|

## Type MS · What encryption is and why we use it

### Answer 4 of 5 · Question 4 part a · 1 mark

2022 JUN · Paper 02 · extract 3 of 6 · find part a in this mark scheme extract

| | | | | |
|---|---|---|---|---|
| 4 (c)(i) | C1 | 5 | | (1) |

## Type MS · What encryption is and why we use it

### Answer 4 of 5 · Question 4 part a · 1 mark

2022 JUN · Paper 02 · extract 4 of 6 · find part a in this mark scheme extract



## Type MS · What encryption is and why we use it

### Answer 4 of 5 · Question 4 part a · 1 mark

2022 JUN · Paper 02 · extract 5 of 6 · find part a in this mark scheme extract

| | | | | |
|---|---|---|---|---|
| 4 (c)(iii) | C3 | Rail (Fence cipher) | | (1) |

## Type MS · What encryption is and why we use it

### Answer 4 of 5 · Question 4 part a · 1 mark

2022 JUN · Paper 02 · extract 6 of 6 · find part a in this mark scheme extract

| Question | mp | Answer | Additional Guidance | Mark |
|---|---|---|---|---|

## Type MS · What encryption is and why we use it

### Answer 5 of 5 · Question 1 part b.i · 1 mark

2023 JUN · Paper 01 · extract 1 of 4 · find part b.i in this mark scheme extract

| | | | |
|---|---|---|---|
| 1(a) | Award **one** mark for any of the following up to a maximum of **two** marks:<br><br>1. Access (shared) peripherals (printers) (1)<br>2. Access the Internet (1)<br>3. Access shared file storage (server, cloud) (1)<br>4. Allows data/files/information to be shared (1)<br>5. Provide communication (email, social media, online meeting, chat) (1)<br>6. Provide collaboration (editing online documents, playing games) (1)<br>7. Allow hot-desking (1)<br>8. Allows centralised software updates/backups (1) | Award any example that can be categorised to a unique reason | 2 |

| Question Number | Answer | Additional Guidance | Mark |
|---|---|---|---|
| 1(b)(i) | **The only correct answer is C**<br><br>***A** is not correct because compression does not encrypt a file*<br><br>***B** is not correct because* encryption does not ensure both sides use the same key<br><br>***D** is not correct because* encryption does not stop hackers accessing a network | | 1 |

## Type MS · What encryption is and why we use it

### Answer 5 of 5 · Question 1 part b.i · 1 mark

2023 JUN · Paper 01 · extract 2 of 4 · find part b.i in this mark scheme extract

| Question Number | Answer | Additional Guidance | Mark |
|---|---|---|---|
| 1(b)(ii) | Award **one** mark for any of the following up to a maximum of **two** marks:<br><br>1. (Electronic/combination) lock on doors (1)<br>2. Swipe/key cards/biometrics (1)<br>3. Security guards/security patrol (1)<br>4. CCTV cameras / surveillance technology (1)<br>5. Asset tagging/marking (1) | | 2 |

| Question Number | Answer | Additional Guidance | Mark |
|---|---|---|---|
| 1(b)(iii) | Award **one** mark for any of the following:<br><br>1. It exploits/relies on human behaviours/tricks people (to reveal sensitive information) (1) | Do not award examples e.g. Pharming without further information. | 1 |

## Type MS · What encryption is and why we use it

### Answer 5 of 5 · Question 1 part b.i · 1 mark

2023 JUN · Paper 01 · extract 3 of 4 · find part b.i in this mark scheme extract

| Question Number | Answer | Additional Guidance | Mark |
|---|---|---|---|
| 1(c) | **The only correct answer is D**<br><br>***A*** *is not correct because 5G does not have a* **high transmission latency**<br><br>***B*** *is not correct because a 5G does not have* ***built-in security***<br><br>***C*** *is not correct because* 5G does not have a **lower bandwidth than 3G and 4G** | | 1 |

| Question Number | Answer | Additional Guidance | Mark |
|---|---|---|---|
| 1(d) | Award **one** mark for:<br><br>• Testing each block of code/subprogram/function independently (as it is written/finished) (1) | Do not accept module in isolation as this is in the question. | 1 |

## Type MS · What encryption is and why we use it

### Answer 5 of 5 · Question 1 part b.i · 1 mark

2023 JUN · Paper 01 · extract 4 of 4 · find part b.i in this mark scheme extract

| Question Number | Answer | Additional Guidance | Mark |
|---|---|---|---|
| 1(e) | Award **one** mark for any of the following up to a maximum of **two** marks:<br><br>1. Belong to/have membership in a professional society (1)<br>2. Attend computer science related conferences/gatherings (1)<br>3. Attaining/gaining training/educational opportunities (1)<br>4. Behave in ethical/legal/moral ways (1)<br>5. Stay up to date with changes in the field/read up-to-date publications (1)<br>6. Use responsible programming practices e.g. due diligence / testing / commenting code for maintainability (1)<br>7. Avoid bias when making design choices (1) | Award examples if they can be categorised to a unique bullet point<br><br>Answers must focus on computer science rather than being generic<br><br>Do not accept things that would be expected as part of an employee's role e.g. writing code | 2 |

| Question Number | Answer | Additional Guidance | Mark |
|---|---|---|---|